

How Do Users Perceive Recommender Systems' Objectives?

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Abstract

Multi-objective recommender systems (MORS) aim to optimize multiple criteria while generating recommendations, such as relevance, novelty, diversity, or exploration. These algorithms are based on the assumption that an operationalization of these criteria (i.e., translating abstract goals into measurable metrics), will reflect how users perceive them. Nevertheless, such beliefs are rarely rigorously evaluated, which can lead to a mismatch between algorithmic goals and user satisfaction. Moreover, if users are allowed to control the RS via their propensities towards such objectives, the misconceptions may further impact users' trust and engagement. To characterize this problem, we conduct a large user study focusing on recommender systems in two domains: books and movies. Part of the study is focused on how users perceive different recommendation objectives, which we compared with well-established metrics aiming at the same objectives. We found that despite such metrics correlating to some extent with users' perceptions, the mapping is far from perfect. Moreover, we also report on conceptual-level differences in users' understanding of RS objectives and how this affects the results. Study data are available from https://osf.io/2n9mf/.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

RS quality perception, Beyond-accuracy, User study

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1 Introduction

Since the early days, the main aim of recommender systems (RS) has been to provide users with accurate recommendations that reflect their known past feedback. Nevertheless, as models have become increasingly accurate, researchers have started to recognize that accuracy alone is not sufficient to ensure high-quality recommendations [55]. In recent years, we have witnessed a paradigm shift, with more and more work addressing perspectives that go *beyond* recommendation *accuracy*. These include notions such as recommendation diversity [24, 36, 73], novelty [12], fairness with respect to different stakeholders [5, 51], and the exploration vs. exploitation trade-off [15, 51]. This shift has led to the emergence of multi-objective recommender systems (MORS) [35], which aim to optimize multiple objectives, including beyond-accuracy ones, while delivering recommendations.

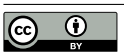
However, any MORS relies on a key assumption: *the metrics used for optimization are aligned with users' perceptions of the corresponding objectives.* If a substantial mismatch exists between perception and metric, the optimization procedure may target a false goal and yield suboptimal results. This issue is even more pronounced in controllable MORS [16, 48], where users can tune the output via an interface. In such cases, failing to reflect users' understanding of the tuning options may result not only in suboptimal performance, but also in a lack of transparency and understandability—two essential properties of recommender models [16, 49]. Despite the potentially serious implications, the problem of describing and quantifying the mismatch between perceived and measured objectives remains understudied [24, 29, 36].

To address this gap, in this paper, we conduct a study focused on user behavior when interacting with controllable MORS. A portion of the study is dedicated to discovering and quantifying the mismatch between users' perceptions of recommendations' relevance and several beyond-accuracy objectives (novelty, diversity, exploration, serendipity, popularity, uniformity, and exploitation) and the corresponding metrics used to quantify these objectives. We analyzed whether users share a common conceptual understanding of these objectives. Then, we focused on the common operationalizations of the corresponding metrics and observed the extent to which each operationalization correlates with users' perceptions. Finally, we analyzed how the conceptual-level mismatch affects the mismatch at the operationalization level.

In particular, our research questions are as follows:

- **RQ1:** How severe is the conceptual mismatch, i.e., do users understand RS objectives differently from how we measure them?
- **RQ2:** How severe is the operationalization mismatch, i.e., how well do different metric implementations align with users' perceptions of the corresponding objectives? In cases with multiple operationalization variants, which provides the best estimate?
- **RQ3:** Does the conceptual mismatch affect the observed misalignment with respect to the metrics' operationalization?

The main results of the study are as follows. Despite users being mostly aware of different RS objectives, they often understand them differently on the conceptual level. On the operationalization level, while the user-perceived RS quality to some extent correlates with



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the corresponding metrics, the mapping is not perfect, and often, metrics aiming at different objectives perform similarly well. Having users who conceptually align with the system's understanding of the objectives helps to significantly reduce this mismatch.

These results provide clear incentives for educating the users about the system's understanding of individual objectives and the need for domain-specific research on metrics operationalization. In addition, our results also reveal that objectives that are inverse to what is commonly understood as beneficial for the user (i.e., uniformity, popularity, and exploitation) may positively affect the overall satisfaction of users. This highlights the need for maintaining a careful balance between individual objectives, as overdosing users with novel, diverse, or explorative recommendations may hurt their satisfaction.

2 Related Work

Our work is connected to several established RS research areas, which we describe in this section.

Multi-objective recommendation. Research in multi-objective recommender systems (MORS) spans various dimensions, as noted by Jannach and Abdollahpouri [35]. MORS can operate at the *aggregate* or *individual* level, with the latter personalizing recommendations for each user. Historically, MORS often employed re-ranking techniques to enhance diversity at the aggregate level [8, 10, 19, 83]. Later works integrated metrics such as fairness [5, 18, 62], novelty [32], and popularity bias [1], alongside individual-level MORS [24, 25, 52–54, 63, 68, 73]. We adopted RLprop [63] as one of the MORS algorithms utilized in this study. Nevertheless, balancing more than two objectives with re-ranking techniques is uncommon [35], while complex settings often rely on Evolutionary Algorithms (EAs) [4, 28, 30, 31, 47, 78, 82]. Algorithms such as NSGA-II [21], NSGA-III [20, 34], and MOEA/D [81] are frequently used, often with custom modifications [28, 30, 31, 78, 82]. Inspired by Hafsa et al. [30], we also employed NSGA-II with a custom utility function tailored to results-level proportionality, while leaving further optimization for future work. While traditional evaluations of RS focus on system-side metrics, HCI research highlights that users often rely on interface cues and cognitive heuristics when interpreting recommendations. Hu and Pu [33] found that organizing recommended items visually (e.g., in grouped categories) significantly enhanced users' perception of diversity. Similarly, Knijnenburg et al. [40] emphasize that transparency, controllability, and user control affect trust and satisfaction. These insights underscore the importance of designing UI elements that not only support interactive control but also clearly signal the intended system objectives, especially in multi-objective settings.

Common objectives and metrics. Over time, numerous objectives have been considered in the context of recommender systems. Besides relevance, *diversity* [9, 72, 79], *novelty* [66, 74], and *exploration* [3, 14, 49]—all used in this study—are among the most common. However, it is not always the case that users desire beyond-accuracy objectives (e.g., they may not actively seek more diverse recommendations); in fact, they may prefer the opposite [27]. Therefore, in our work, we also incorporated inverse formulations of these objectives: *uniformity*, *popularity*, and *exploitation*.

Naturally, there are many ways to measure each of these objectives [3, 9, 73, 74], or combinations thereof [17, 60]. From this broad set, we focused on metrics that are single-purpose (i.e., designed to capture a single objective), widely adopted, and applicable in our target domains. Many of these metrics rely on an underlying notion of item-item distance. To operationalize this, we used a diverse set of item distance matrices, selected based on the availability of suitable data in both evaluated domains. In particular, following [24], we evaluated collaborative, metadata-based, and content-based distance variants, whenever such interpretations were meaningful for the given objective.

Results-level proportionality in MORS. Traditional MORS may not align recommendations with the relative importance of objectives (i.e., user's propensities towards individual objectives). Peska and Dokoupil [63] introduced *results-level proportionality*, requiring MORS to deliver high-quality recommendations proportionally to propensity scores. The RLprop algorithm [63] implements greedy re-ranking to achieve proportionality across objectives, incrementally building recommendation lists. Several derivates of the RLprop algorithm have been proposed in [25], modifying both the backbone (using either simplified item-wise greedy selection, or an evolutionary algorithm) and the interpretation of the users' propensity feedback. Both the original RLprop and the same derivates as proposed in [25] are used as MORS variants in this work.

Existing propensity estimation methods [22, 27, 37] involve trade-offs between user-selected and available items. Nevertheless, recent work in user modeling goes beyond observable behavior to capture users' conceptual preferences and expectations. Nguyen et al. [58] demonstrate that users vary in their appetite for diversity, novelty, and serendipity, and that personality traits can predict such preferences. Incorporating such user-level modeling into RS can help align recommendations with individuals' subjective perceptions of value, not just their interaction histories. This perspective supports the idea that controllable MORS should integrate adaptive mechanisms that learn and respect users' conceptual goals over time. Our method for estimating users' propensity follows from [22, 27, 37], while we also adopt an interface allowing users to self-declare their propensities toward individual objectives, i.e., to actively control what kind of recommendations they receive.

Controlling recommender systems. While traditional systems optimize recommendations passively, user-controllable interfaces can enhance recommendation acceptance [7, 32, 39, 59, 70]. Proposed interactive designs include SetFusion [61], visual controls [56], and sliders [32, 51, 70]. Early systems such as PeerChooser [59] and TasteWeights [7] introduced visual control metaphors. More recent work explores not only slider-based or contextual controls [56, 70], but also how such interfaces can be personalized based on user characteristics and goals, as in music genre exploration [48]. Recent studies [23, 25, 27] have explored propensity-based controls, but noted interpretability challenges. A complementary direction to system-side modeling is participatory design; engaging users in shaping how RS operate and optimize. Verbert et al. [75] argue that value-sensitive design is crucial for aligning RS behavior with users' goals, particularly when beyond-accuracy objectives like fairness or diversity are involved. Participatory methods allow users to express not only their preferences, but their understanding

and priorities among competing objectives, offering an avenue for reducing perception mismatches and increasing system legitimacy.

Objectives perception. Aligning the interpretation of individual objectives between users and the system is a crucial aspect of controllable MORS design, but it can also be relevant in other scenarios. Nevertheless, studies explicitly addressing this mismatch remain rare [24, 29, 36, 79]. Users' subjective interpretations of objectives can differ significantly from system definitions. Ekstrand et al. [29] observed perception gaps even for familiar qualities like novelty and diversity, while the gaps in diversity perception were confirmed by the work of Dokoupil et al. [24] as well. Jesse et al. [36] and Wang and Chen [76] showed that these gaps are influenced by implementation details and user traits. User studies also highlight how diversification affects decision difficulty and satisfaction [79], and how fairness perceptions depend on calibration and exposure biases [2]. These findings support the need for conceptual clarity and user-aware evaluation.

Conceptually closest to our current work is probably Ekstrand et al. [29], who compared the metrics of diversity, novelty, and accuracy with how users perceived these aspects in recommendations. However, they considered only one metric per objective and provided users with recommendations optimized solely for relevance—conditions that may have influenced the observed results. In contrast, we incorporated additional objectives as well as multiple operationalizations for individual objectives. Moreover, we provided users with both relevance-oriented and controllable multi-objective recommendations.

Explainability and Socio-technical framing of algorithmic objectives. Recommender objectives like diversity, fairness, or popularity are not just technical preferences; they reflect normative choices about what counts as a "good" outcome. Socio-technical critiques have questioned overemphasis on accuracy [55] and called for fairness-aware systems [5]. These works provide a conceptual basis for the adoption of MORS. Nevertheless, Binns et al. [6] argue that users often question the legitimacy of algorithmic decisions, particularly when they cannot see how outcomes were derived. Similarly, Lee and Baykal [45] show that algorithmic group assignments are perceived as less fair than human decisions unless accompanied by transparent reasoning. These findings suggest that when RS optimize for latent objectives without surfacing their logic, users may experience the system as opaque or unfair. Framing RS design as a socio-technical problem, where metrics encode implicit values, can help address perception mismatches and build more accountable systems.

Meanwhile, explainability research has shown that users often form mental models of recommendation logic that diverge from the actual system behavior [57, 67]. Kunkel et al. [44] showed that opaque recommenders erode trust, while personalized explanations enhance satisfaction and acceptance. Tintarev and Masthoff [71] categorize explanation goals (including transparency, trust, and scrutability), many of which are especially relevant when users can tune recommendation objectives themselves. In multi-objective RS, failing to explain the meaning and trade-offs behind objectives can contribute to perception–metric mismatches, suggesting a need for contextualized, multi-criteria explanations. From this perspective, the propensity controlling interface can also contribute to the explainability of the RS outcome.

3 Models and Methods

3.1 Notation

We reserve lower-case u, v for users and i, j for items, while $\mathcal{U}$ and $\mathcal{I}$ denote the sets of all available users and items, respectively. We further denote the list of recommendations as L, existing user feedback as $r_{u,i} \in \mathbf{R}$, and estimated relevance as $\hat{r}_{u,i} \in \hat{\mathbf{R}}$. Several metrics utilize a matrix of item-item distances, denoted as $\mathbf{D}$, where the subscript (e.g., $\mathbf{D}_{CF}$) indicates a particular variant of the distance calculation. Finally, the user's history, i.e., the items with which a user previously interacted, is denoted as $\mathcal{H}_u$.

3.2 Datasets and Pre-processing

For the experimental evaluation, we used the same datasets and pre-processing steps as described in [24]. In particular, we focused on the movie and book domains, utilizing the *MovieLens Tag Genome 2021* [42] and *Goodbooks-10k* [80] datasets, respectively. Both datasets were filtered to only include items with a valid cover image and metadata. For Goodbooks, we kept only the most relevant genre per book. For MovieLens, we removed movies released before 1985, users who had not rated any movies released since 2016, and movies with fewer than 10 remaining ratings. After pre-processing, the dataset sizes were $|\mathcal{U}| = 53K$, $|\mathcal{I}| = 6.5K$, $|\mathbf{R}| = 4.6M$ for Goodbooks, and $|\mathcal{U}| = 18K$, $|\mathcal{I}| = 16K$, $|\mathbf{R}| = 5.1M$ for MovieLens.

3.3 Recommendation Objectives and Metrics

In this study, we incorporated *relevance*, *diversity*, *novelty*, *exploration*, *uniformity*, *popularity*, *exploitation*, and *serendipity* objectives. For each objective, we focused on well-known operationalization variants frequently used in results tuning and/or evaluation. In many cases, these include meta-metrics that operate on top of arbitrary distance functions. We adopt three such distance variants based on collaborative, content, and metadata information. Specifically, we used cosine distance between user rating vectors as the *collaborative* distance (*CF*), cosine distance between vector representations of plot summaries as the *content-based* distance (*CB*)[1], and cosine distance between genre vectors as the *genre-wise* distance (*G*). We now describe the operationalization variants used for each objective.

Relevance. We define the estimated relevance *REL* of the list L for a user u as the mean of the estimated relevance scores of the recommended items. Additionally, we introduce a normalized estimated relevance REL_N, where item scores are transformed with respect to the empirical cumulative distribution trained on all estimated ratings of the current user. This effectively captures the ranking correctness of the list. The main motivation for this per-user normalization is that many recommendation algorithms (including $EASE^R$ [69], used in this study) yield widely varying score ranges

[1] We used the OpenCLIP implementation of the CLIP [65] model, 'ViT-B-32' variant, pre-trained on the 'laion2b_s34b_b79k' dataset (https://huggingface.co/laion/CLIP-ViT-B-32-laion2B-s34B-b79K).

across users, which may hinder comparison with user-perceived relevance.

Diversity and Uniformity. The best-known diversity metametric is intra-list diversity (ILD, [83]), defined as the mean distance between all recommended items:

$$ILD(L, \mathbf{D}) = \sum_{i,j \in L, i \neq j} \frac{\mathbf{D}_{i,j}}{|L| * (|L| - 1)} \tag{1}$$

where $\mathbf{D} \in \mathbb{R}^{n \times n}$ is an arbitrary item-item distance matrix. In our work, we used collaborative ($\mathbf{D}_{CF}$), content-based ($\mathbf{D}_{CB}$), and genre-wise ($\mathbf{D}_{G}$) variants. In addition to ILD, we also used the genre-specific binomial diversity (*BinDiv* [73]), which combines genre *coverage* and *non-redundancy* in L. To keep performance reasonable, we used only the global component of genre relevance (i.e., $\alpha = 0$ in Eq. (8) of [73]).

We define uniformity as the inverse of diversity. For each diversity metric, we define a corresponding uniformity metric as its complement, e.g., $UNF_{CF} = 1 - ILD_{CF}$.

Novelty and Popularity. Following Vargas and Castells [74], we define the novelty of an item as the complement of its popularity, $NOV(i) = 1 - POP(i)$, where popularity is the fraction of users who interacted with item i: $POP(i) = |\{v; \mathbf{R}_{v,i} > 0\}|/|\mathcal{I}|$. The novelty and popularity of a list are computed as the mean novelty/popularity of its items.

Exploration and Exploitation. Conceptually, exploration measures the extent to which the recommended items deviate from a user's known interests. In this study, we operationalize exploration as the unexpectedness [3] of the recommended list relative to the user's historical feedback:[2]

$$EXPLR(u, L, \mathbf{D}) = \sum_{i \in L, j \in \mathcal{H}_u} \frac{\mathbf{D}_{i,j}}{|L| \cdot |\mathcal{H}_u|} \tag{2}$$

Exploitation, by contrast, measures the degree to which the recommendations align with a user's known preferences, and is defined as the complement: $EXPLT(u, L, \mathbf{D}) = 1 - EXPLR(u, L, \mathbf{D})$.

Serendipity. Beyond directly controllable objectives, we also examine perceived and measurable serendipity. We adopt the definition $serendipity(u, i) = relevance(u, i) \times unexpectedness(u, i)$ [38]. For relevance, we use either the raw estimated score $REL(u, i)$ or the normalized version $REL_N(u, i)$, denoted as SER_{REL} and SER_{REL_N}, respectively. Alternatively, we define a feedback-based metric SER_{FB} using binary user feedback ($relevance(u, i) = 1$ for clicked items, 0 otherwise). For unexpectedness, we use the item-specific form of exploration: $unexpectedness(u, i) = EXPLR(u, i, \mathbf{D})$. List-level serendipity is computed as the mean serendipity across all recommended items. In addition to the above-mentioned approach, we also utilized two unexpectedness variants from literature: pointwise mutual information SER_{pmi} and a content-based surprise SER_{JC} as defined in [38]. These were, as per their definition, only combined with feedback-based relevance.

[2] I.e., our definition is a special case of Eq. (14) in [3].

3.4 Recommendation Algorithms

We employed one variant of a single-objective recommender system (SORS) and several variants of multi-objective recommenders (MORS). As for SORS, we used $EASE^R$ [69], a prominent representative of linear autoencoders. We pre-trained $EASE^R$ on each dataset using L2 regularization $\lambda = 0.01$ and considered all items with which a user had previously interacted[3] as part of their history.

For MORS, we used the estimated scores from $EASE^R$ as the basis for the *relevance* objective and incorporated additional objectives: *novelty*, *diversity*, and *exploration*. We defined these objectives as follows:

- Mean estimated rating $\sum_{\forall i \in L} \hat{r}_{u,i}/|L|$ for *relevance*.
- Mean complement of item popularity $\sum_{\forall i \in L} NOV(i)/|L|$ for *novelty*.
- Collaborative intra-list diversity $ILD(L, \mathbf{D}_{CF})$ for *diversity*.
- Collaborative unexpectedness $EXPLR(u, L, \mathbf{D}_{CF})$ for *exploration*.

All MORS variants aimed to maintain results-level proportionality [64], i.e., maximizing all objectives while preserving their relative importance as defined by the user through a dedicated interface.[4] To maintain proportionality, the individual objectives needed to be commensurable, so we transformed them using the empirical cumulative distribution function (CDF). All MORS algorithms were derivatives of the RLprop algorithm [64], differing in how they interpreted user feedback and which optimizer was used (item-wise, greedy list-wise, or evolutionary). Since a detailed comparison of MORS variants is beyond the scope of this paper, we omit the specifics for space reasons. Full implementation details are available in the online supplementary materials.

4 Experimental Design

The study was conducted in the movie and book domains using the EasyStudy framework [26]. The study (see Figure 1, top) comprised a pre-study phase with a *task description*, *informed consent*, *pre-study questionnaire*, *preference elicitation*, and *list-level diversity assessment*. The main phase interleaved *recommendation blocks* with *after-block questionnaires*, and the study concluded with a *post-study questionnaire*.

Note that while our institute's regulations do not require formal ethical review for such studies, we adhered strictly to ethical guidelines for human subjects research, including the principles of the Belmont Report and Prolific's ethical research standards. Specifically, all participants took part voluntarily, received a detailed study description, were informed of their right to withdraw, and were provided with the estimated study duration in advance. They also gave informed consent for the publication of anonymized data. No personally identifiable information was collected, and to the best of our assessment, the study posed no potential risk to participants.

[3] I.e., items selected during preference elicitation and those chosen in previous recommendation blocks.

[4] Users could adjust their propensities toward individual objectives after each recommendation iteration. For the first iteration, propensities were estimated using the method from [24], based on selections made during preference elicitation.

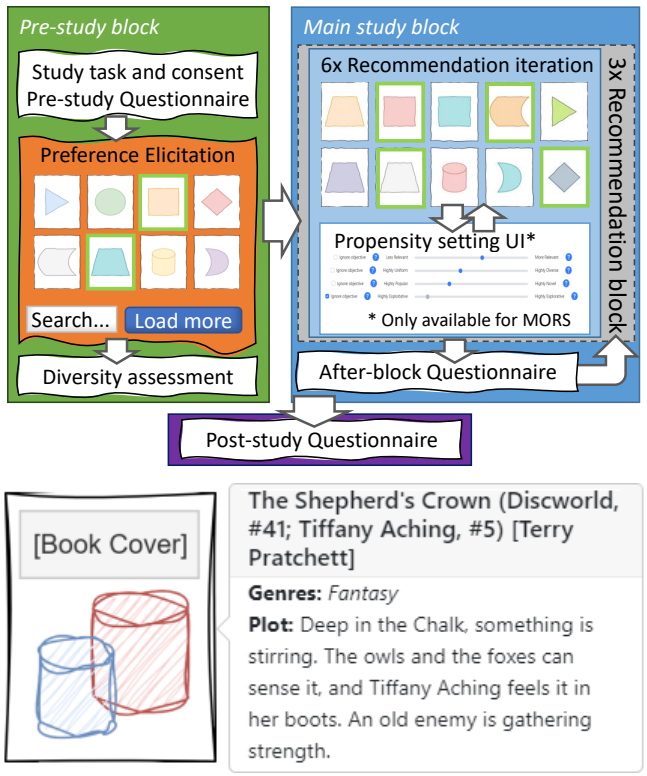


Figure 1: Experimental design: Study overview (top) and recommended item visualization on mouse hover (bottom). Note that actual movie posters / book covers were displayed to users.

4.1 Study Flow

The high-level study flow is depicted in Figure 1 (top). Below, we provide a detailed description of the study components relevant to this paper, i.e., the *pre-study questionnaire*, *preference elicitation*, *recommendation blocks*, and *after-block questionnaires*. A commented study walkthrough is available in the online repository.

Pre-study. Prior to beginning the study, users were presented with a mission statement, detailed instructions, and were asked to provide informed consent for the publication of anonymized data. They were then routed to a pre-study questionnaire focused on their familiarity with the domain, recommender systems, and their conceptual understanding of objective criteria.

In PQ1, we asked whether users knew what recommender systems (RS) are and how they work. In PQ2–6, we inquired whether users had previously heard the terms "relevance", "diversity", "novelty", "exploration", and "exploitation" (i.e., familiarity) in the context of RS. In PQ7–11, we asked about users' conceptual understanding of these terms. For PQ7 (relevance), users could select from: *match to existing interest*, *items for immediate utilization*, *understand the variety of the offer*, and *finding unexpected yet interesting items*.[5] For PQ8–11, the same options were provided, corresponding to the system's interpretations of *exploration*, *diversity*, *novelty*, and *exploitation*: *items different from already consumed*, *heterogeneous items*, *niche items not generally known*, and *items similar to already consumed*, respectively. In all cases, users could also provide a free-text response. Finally, PQ12 asked about users' content consumption frequency in the target domain.

Preference Elicitation. To provide realistic recommendations, we first solicited users' preferences using the method described in [24]. Specifically, users were shown a list of items and asked to indicate which ones they *had previously experienced and liked* (i.e., positive-only feedback). The prompts were sampled to include both popular and lesser-known items and to ensure sufficient diversity.

Users were required to select at least five items before proceeding to the next step. To support this, they could request additional items or manually search for specific ones.

Recommendation Blocks. In this step, users received a sequence of 18 recommendation iterations, organized into three *blocks* of six iterations each. Each iteration included 10 recommended items, and all recommendations within a single block were served by the same recommendation algorithm. Participants were asked to select the items they would *consider watching or reading next*. Note that within a single block, each item could be recommended at most once, and recommendation algorithms could use the user's selections from previous iterations to update their profile.

In the case of SORS-based recommendations, users had no means to tune the system, whereas in the case of MORS, users could update their propensities toward individual objectives (*relevance*, *novelty*, etc.) after each iteration via a dedicated GUI (see Figure 1). After the final iteration of each block, users were directed to complete an after-block questionnaire.

After-block Questionnaire. In the after-block questionnaire, users were asked to reflect on the preceding set of recommendation iterations. In particular, they were asked whether the recommendations *matched their interests* (AQ1, relevance), were *mostly novel* (AQ2, novelty), *highly different from each other* (AQ3, diversity), *unexpected yet interesting* (AQ4, serendipity), *different from their usual choices* (AQ5, exploration), *similar to their usual choices* (AQ6, exploitation), *mostly popular* (AQ7, popularity), and *mostly similar to each other* (AQ8, uniformity). Finally, we asked whether they were *overall satisfied with the recommendations* (AQ9). Responses were collected on a 6-point Likert scale (with no mid-point),[6] and users could also select an "I don't understand" option if needed.

4.2 Study Context

During the study, we recorded all user actions, primarily the selection of items and responses to the questionnaire items. Each displayed item was represented by a cover image, with additional information (title, genres, plot description) shown on mouse hover—see Figure 1 (bottom).

To ensure usability, a minimal screen resolution was required so that all items would be visible at once. Four attention checks were included in the questionnaires (both nonsensical and instruction-based), and participants who failed more than one check were excluded from the analysis.

[5]Note that full texts of questionnaire questions and answers are also available from the online supplementary materials: *https://osf.io/2n9mf/*.

[6]In evaluation, we assign a numeric value from the $[-3, 3]$ interval to each answer. For example, "Strongly disagree" corresponds to -3, while "Agree" corresponds to 2.

Note also that while brief instructions were always visible at each step, we provided an expandable sidebar with further details, accessible via a dedicated "Instructions" button.

5 Results

A total of 279 participants started the user study, of whom 229 completed it. We omitted 21 participants who failed attention checks, resulting in a final sample size of 208 participants. Most participants (195) were sourced from Prolific.com, while the remainder were recruited through the authors' networks. All participants were prescreened for fluent English, and Prolific users were additionally required to have at least a 99% approval rate and a minimum of 10 prior submissions. The average completion time was 27 minutes, with a median of 24 minutes.

Participants were reasonably distributed with respect to gender (50% male, 49% female, and 1% other/unknown) and age (min: 19, max: 59, mean: 28 years). Most participants were from Europe (71%), followed by North America and South Africa (11% each). In total, participants made over sixteen thousand item selections. On average, users selected 10 items during preference elicitation and 4 items per recommendation iteration. Additionally, most participants reported regularly consuming content in the target domains (77% for movies and 85% for books), which aligns well with the study design, as these users are likely to encounter movie or book recommendations in real life.

5.1 RQ1: Conceptual-level Mismatch

In the pre-study questionnaire, over half of the participants (57%) reported having little knowledge of recommender systems, while another 17% reported no knowledge at all. Despite this, most participants had previously encountered the terms "relevance" (87%), "diversity" (73%), "novelty" (64%), and "exploration" (66%) in the context of recommender systems. Only for "exploitation" was the negative response more frequent (66%).

In the second part of the questionnaire (PQ7–PQ11), we examined to what extent the operationalized definitions of recommendation objectives aligned with participants' conceptual understanding (see Figure 2 for details). The interpretation of relevance was largely consistent with its operationalization (88%). However, consistency was considerably lower for beyond-accuracy objectives. Notably, over 30% of participants interpreted *diversity* and *novelty* as *exploration*. For *exploration* itself, responses were even more fragmented: 38% interpreted it as exploration, another 38% as novelty, and 22% as diversity. In contrast, 79% of participants interpreted *exploitation* as expected.

Key Observation. *While users tend to be familiar with beyond-accuracy objective criteria, they often interpret them differently from how the corresponding metrics are defined.*

This observation is partly supported by Wang and Chen [77], who found that users' perception of beyond-accuracy metrics may be influenced by their personality traits. In any case, due to the substantial conceptual mismatches observed in the interpretation of *diversity*, *novelty*, and *exploration*, we now investigate how these mismatches affect the following research questions.

5.2 RQ2: Operationalization-level Mismatch

In this research question, we aim to quantify the degree of mismatch between how individual recommender system objectives are measured and how users perceive them. To this end, we analyzed the responses from the after-block questionnaires and compared them with the corresponding metric values for each recommendation iteration.[7]

First, we describe the overall questionnaire results. Figure 3 shows the mean responses grouped by domain and recommendation algorithm. In general, MORS variants were rated lower in terms of perceived relevance (AQ1; 1.95 vs. 0.93), but significantly higher on beyond-accuracy criteria (AQ2, AQ3, and AQ5), with an average difference of about one scale point (t-test $p < 3.8e{-}14$). This pattern aligns with the metric-based evaluation.

More importantly, MORS variants also significantly outperformed SORS in terms of serendipity (AQ4; 0.41 vs. 0.09; $p = 0.019$), particularly in the movies dataset. This may be due to the book domain being less familiar to users, making unexpected items easier to find. The strongest dataset-related difference was observed for novelty (AQ2), which was significantly higher in the books dataset (0.46 vs. -0.02; $p = 0.001$), supporting the above hypothesis.

To evaluate the mismatch between metrics and users' perception of recommendation objectives, we analyzed Pearson's correlations between system-side metrics and questionnaire responses. Figure 4 shows the pairwise results. We observe that all metric variants of *relevance*, *novelty*, *diversity*, and *exploration* correlate positively with the corresponding questionnaire responses. However, the strength of correlation varies widely—from very weak (0.15 for *REL* and AQ1 in the movies domain) to fairly strong (0.55 for $EXPLR_{CF}$ and AQ5 in movies).

The alignment between different operationalizations of the same objective also varies substantially, where depending on the dataset, different metrics perform the best. Consider, for example, perceived diversity (AQ3): in the book domain, genre-based metrics (BinDiv and ILD_G) showed the strongest correlation with AQ3, whereas in the movie domain, the best-performing metric was ILD_CF. Statistical tests[8] confirm these findings. In the book domain, both BinDiv and ILD_G significantly outperformed ILD_CB and ILD_CF (p-values $\in [0.015, 0.041]$). In the movie domain, we observed a significant difference between ILD_CF and ILD_CB (p-value: 0.013), and a marginally significant difference between ILD_CF and BinDiv (p-value: 0.059).

Similar patterns were found for other objectives. For AQ5 (exploration), EXPLR_G significantly outperformed EXPLR_CF in the book domain (p-value: 0.005), while in the movie domain, EXPLR_CF significantly outperformed both EXPLR_CB (p-value: 0.004) and EXPLR_G (p-value: 0.043). For AQ6 (exploitation), EXPLT_G significantly outperformed EXPLT_CF in books (p-value: 0.018), whereas EXPLT_CF performed better than EXPLT_CB in movies (p-value: 0.036). For AQ8 (uniformity), no significant differences were observed in the book domain, but in movies, both UNF_CF and UNF_G significantly outperformed UNF_CB (p-value ≤ 0.006).

Overall, these findings show a consistent pattern: $\mathbf{D}_G$ is generally more effective in the book domain, $\mathbf{D}_{CF}$ performs best in the movie

[7] Specifically, we used the mean values of the metrics computed on the individual recommendation lists.
[8] Applying Fisher's z-transformation and one-sided z-test.

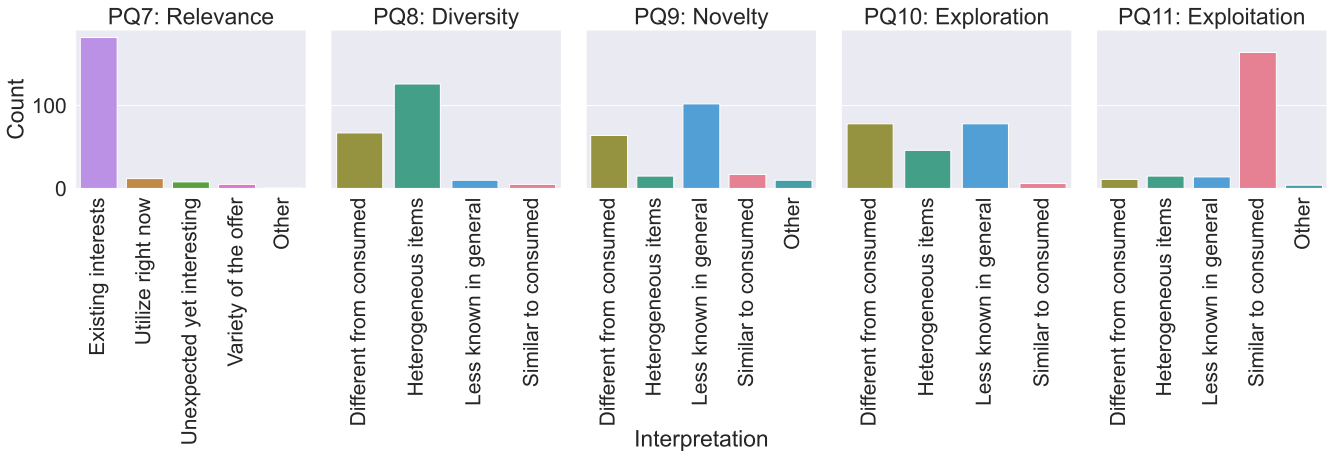


Figure 2: Distribution of responses on the conceptual understanding of recommender systems' objectives. Note that full texts of both questions and possible answers are available from the study's repository.

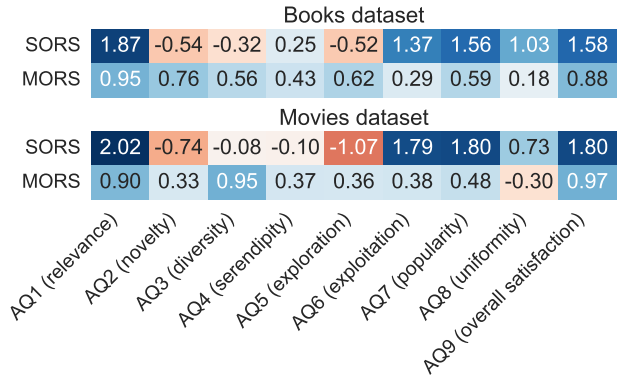


Figure 3: Mean answers for after-block questionnaire, grouped by dataset and recommendation algorithm. Note that Likert scale responses were mapped to $[-3, 3]$ interval, where higher values correspond to higher agreement with the statement.

domain, and $\mathbf{D}_{CB}$ is inferior in both. This supports and extends the results of Jesse et al. [36], who found that the correspondence between measured and perceived recommendation diversity heavily depends on the feature space used to compute it.

Key Observation. *Users' perception of RS qualities is, to some extent, aligned with corresponding metrics. However, the strength of this alignment varies significantly, and different metrics perform best for different datasets.*

Regarding estimated relevance, we find that per-user normalization adds substantial value: REL_N outperforms REL in both datasets (p-value ≤ 0.007), motivating further investigation into alternative normalization strategies in future work.

Interestingly, when looking at questionnaire items related to beyond-accuracy objectives (AQ2, AQ3, and AQ5), we observe that other beyond-accuracy metrics are often equally or even more strongly aligned with user perceptions than the metrics specifically designed to capture the corresponding quality. For example, EXPLR_CF is strongly correlated with perceived exploration (AQ5) in movies ($\rho = 0.55$), but so are NOV and ILD_CF. Similar cross-objective overlaps appear for all perceived beyond-accuracy criteria, inverse criteria (AQ6–AQ8), and even relevance (AQ1).

Part of this may be explained by strong correlations between some of the metrics themselves—particularly those relying on the same distance matrices. However, there are notable exceptions. For instance, in the movie domain, both ILD_CF and EXPLR_G show equal correlation with AQ3 (diversity), yet their mutual correlation is only $\rho = 0.55$.

Key Observation. *Metrics often exhibit equal or better alignment with additional objectives they were not specifically designed to capture.*

Let us now turn our attention to serendipity. Our results confirm the widely held belief that serendipity is a particularly difficult concept for users to grasp [41, 43]. The correlations between AQ4 (serendipity) and all evaluated metrics were negligible ($\rho < 0.2$), with the exception of a few serendipity-specific metrics. Even in those cases, the correlations were weak: $\rho = 0.24$ for $SER_{FB,G}$ in the books dataset and $\rho = 0.27$ for $SER_{FB,CF}$ in the movies dataset were the highest observed values.

Notably, none of the serendipity metrics based on estimated relevance yielded non-negligible correlations. Since the SER_{FB} family relies on actual user feedback, its applicability is limited in tasks such as recommendation post-processing or model fine-tuning. These findings highlight the need for developing more actionable and precise metrics for measuring serendipity.

Finally, although not directly tied to a specific objective, we examined how users' overall satisfaction (AQ9) relates to various RS metrics. Interestingly, satisfaction correlated most strongly with

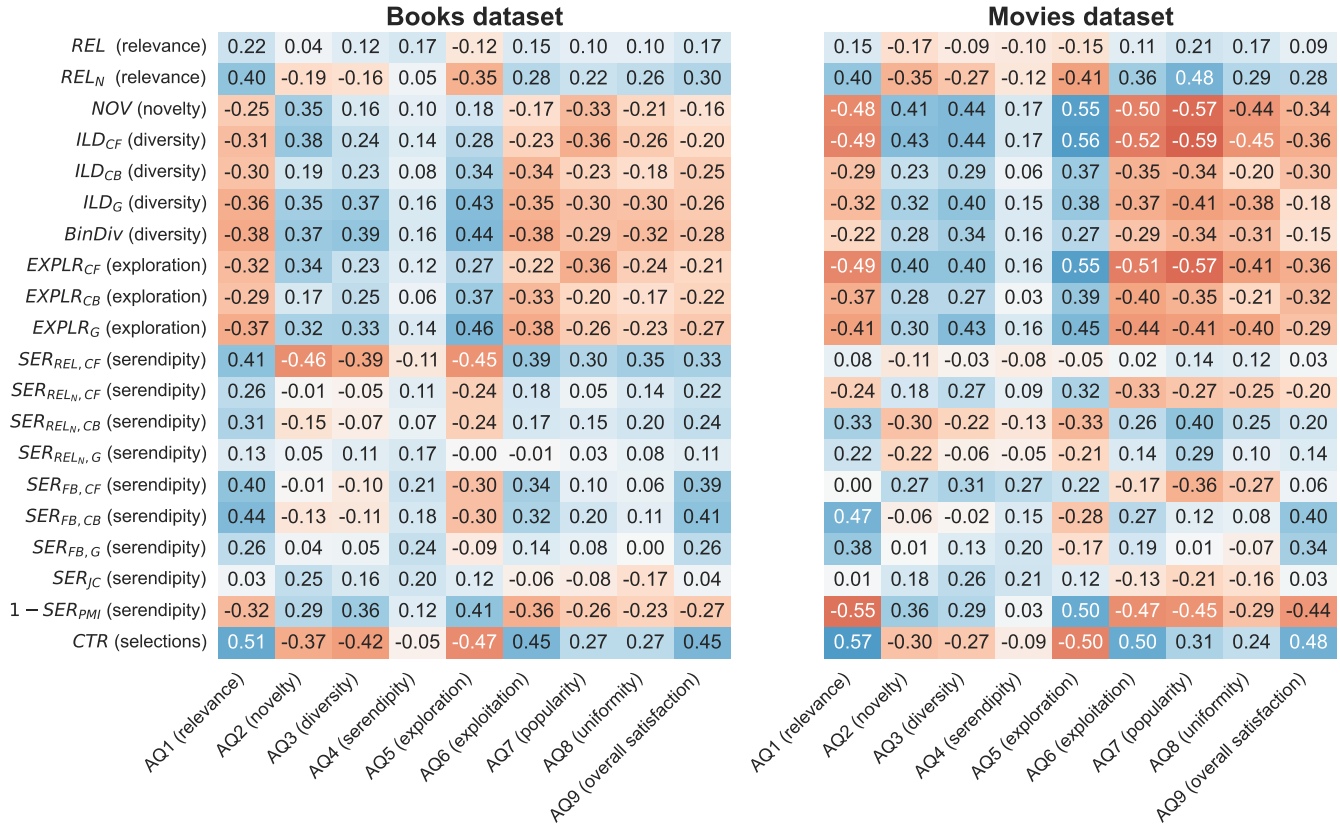


Figure 4: Pearson's correlation between RS quality metrics and users' quality perception (i.e., after-block questionnaire results).

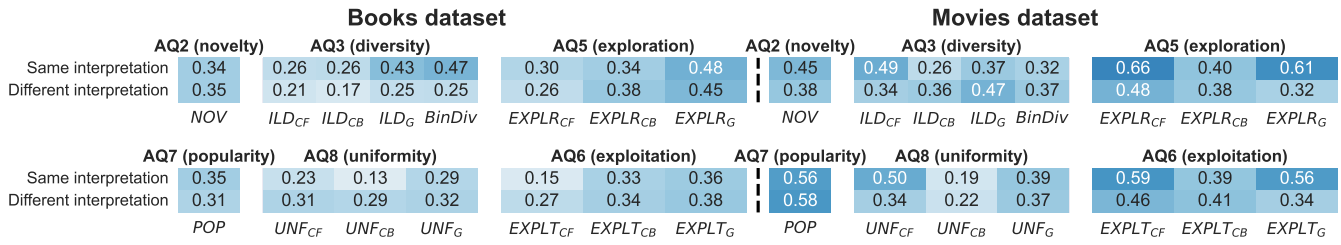


Figure 5: Differences between Pearson's correlations for subgroups of users who shared the same conceptual interpretation of *novelty*, *diversity*, and *exploration* as the system, and those who did not.

users' actual selection behavior (CTR), as well as with estimated relevance (REL_N), popularity, uniformity, and exploitation metrics.[9] In contrast, all evaluated metrics for novelty, diversity, and exploration showed negative correlations with overall satisfaction.

These findings contrast with several earlier studies [11, 23, 29] that reported positive effects of beyond-accuracy objectives. However, closer inspection reveals that those studies either did not specifically optimize for such objectives [29], or reported only moderate effects [11, 23]. Our results emphasize the need for carefully tuning the intensity of beyond-accuracy objectives, as their overuse may negatively impact user satisfaction. While some variants of SER_{REL} and SER_{REL_N} correlated positively with satisfaction, their effects did not surpass those of relevance or inverse metrics. Although SER_{FB} showed somewhat stronger correlations, its reliance on actual user feedback limits its applicability beyond evaluation scenarios.

5.3 RQ3: Impact of Conceptual-level Mismatch on Operationalization-level Mismatch

One may wonder to what extent the operationalization mismatch could be reduced if users and the system conceptually agree on

[9] These metrics were omitted from Figure 4 due to space constraints, but can be derived via the duality of their definitions (e.g., $UNF_{CF} = 1 - \text{ILD}_{\text{CF}}$) by inverting the corresponding beyond-accuracy metrics.

the meaning of individual objectives. To evaluate this, we clustered study participants according to their interpretation of the objectives (PQ8–PQ10). Specifically, for each of *diversity*, *novelty*, and *exploration*, we identified two subgroups of users: those whose conceptual interpretation of the metric aligned with the system's, and those whose interpretation differed. We then computed Pearson's correlations separately for each subgroup. Figure 5 (top) presents the results.

In several cases, metrics that already exhibited strong alignment with user perception (i.e., high correlation in Figure 4) showed further improvement for users whose conceptual understanding matched that of the system. We focused on these improved metric–objective pairs and tested whether the differences were statistically significant.

In the book domain, we observed significant improvements in AQ3 (diversity) for both ILD_G (p-value: 0.041) and BinDiv (p-value: 0.018). In the movie domain, AQ3 showed a marginally significant improvement for ILD_{CF} (p-value: 0.052)[10], and AQ5 (exploration) showed significant improvements for both $EXPLR_{CF}$ (p-value: 0.008) and $EXPLR_G$ (p-value: 0.0006).

Key Observation. *When users conceptually align with the system's interpretation of RS objectives, the operationalization mismatch decreases for several diversity and exploration metric variants.*

We also examined whether the duality assumption holds for inverse metrics, i.e., whether the same subgroup definitions used for *novelty*, *diversity*, and *exploration* could also be applied to their counterparts: *popularity*, *uniformity*, and *exploitation*. In these cases, we observed significant improvements in the movies dataset for AQ8 (uniformity) and AQ6 (exploitation). However, this effect did not hold in the books dataset. As a result, we consider these findings to be mixed and in need of further investigation.

6 Conclusions and Limitations

Understanding how users perceive individual RS objectives is a crucial step toward developing next-generation multi-objective recommender systems (MORS) that allow users to customize their outputs. In this work, we identified common misconceptions and measured the degree of operationalization mismatch for several widely used RS objectives and their associated metrics. While there is some alignment between metric-based optimization and user perception, notable discrepancies remain. Furthermore, we observed that metrics targeting different objectives often overlap and may sometimes better capture users' perceptions than those designed specifically for a given objective. We also showed that when users and the system conceptually agree on the meaning of an objective, the mismatch is substantially reduced for several metrics.

Based on our findings, we offer four key recommendations for future research: **(i)** It is essential to validate that the operationalizations of individual objectives align with how users actually perceive them in a given domain. There is also room for developing improved operationalization variants or adaptive approaches tailored to specific (e.g., long-term) users. **(ii)** There appears to be added value in informing or educating users about how the system interprets individual objectives—particularly in systems that allow user control over them. **(iii)** In offline evaluation scenarios, researchers should avoid overstating small differences in beyond-accuracy metrics, given the substantial gap between metric-based evaluations and user perception. **(iv)** When beyond-accuracy objectives are emphasized too heavily, users may actually prefer recommendations that emphasize popularity, uniformity, and proximity to their known preferences.

The current study has several limitations that can be addressed in future work. The most notable is that users' perceptions of RS objectives were collected only after several recommendation iterations. This introduces volatility in interpreting the performance of the recommendation algorithms across iterations. We opted to compute simple averages, but more sophisticated aggregation approaches could be explored in future research.

Additionally, each perceived objective was assessed using a single questionnaire item. While this design choice helped limit the cognitive burden on participants—given the number of objectives studied—it may have introduced interpretation bias. Our focus was on achieving broad coverage across many objectives, rather than conducting an in-depth examination of any single one.

Since this study was intended as exploratory, we did not construct an explicit causal or dependency model (as in [13, 29, 50]), but instead provided one-to-one comparisons that could serve as a basis for such models in future work.

The choice of metrics was based on those most commonly used in the literature, but many other variants exist. In the future work, we especially plan to focus on metrics that aim to incorporate multiple objectives, such as diversity and relevance [46, 60]. Importantly, the dataset collected in this study supports post-hoc evaluation of alternative metrics, which we plan to pursue.

Finally, while the sample size was sufficient for the general-level analyses presented here, increasing the number of participants would enable finer-grained subgroup analyses—for example, clustering users based on shared interpretations in the pre-study questionnaire.

[10] However, somewhat contradictory results were observed for ILD_G, where correlation was higher for users with different interpretations. The difference was not significant (p-value $\sim$ 0.14), but remains surprising.

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